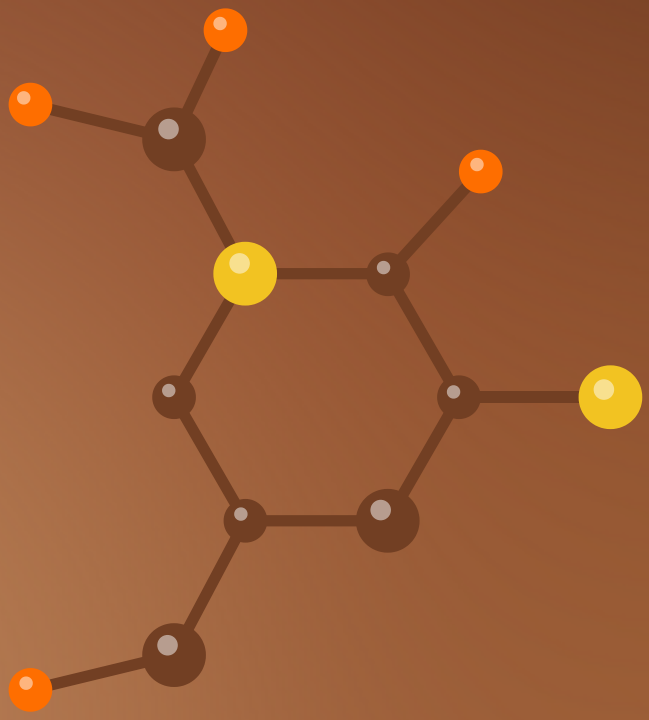
ALLEN CAREER INSTITUTE
path to success
KOTA (RAJASTHAN)



ORGANIC CHEMISTRY

JEE (MAIN+ADVANCED)

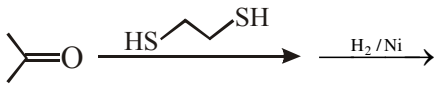
ENTHUSIAST COURSE

EXERCISE

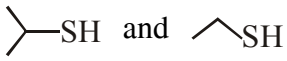
Reduction

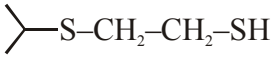
English Medium

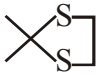
EXERCISE # O-1

1. 

The end products of the reaction are :

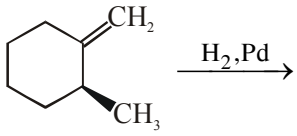
(A)  and

(B) 

(C)  and H_2O

(D) 

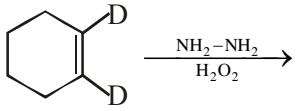
RD0001

2. 

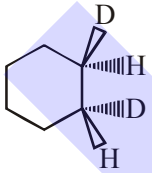
Products of the above reaction will be :

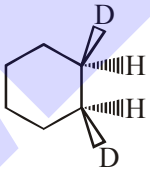
(A) Racemic mixture (B) Diastereomers (C) Meso (D) Structural isomer

RD0002

3.  Product :

(A) 

(B) 

(C) 

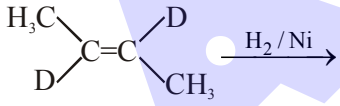
(D) Both (B) and (C)

RD0003

4. On catalytic reduction with H_2/Pt how many alkenes will give n-butane ?

(A) 1 (B) 2 (C) 3 (D) 4

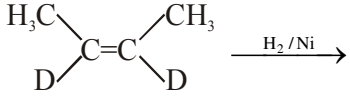
RD0004

5. 

Product of above reaction will be :

(A) Racemic mixture (B) Diastereomers (C) Meso (D) Constitutional isomers

RD0005

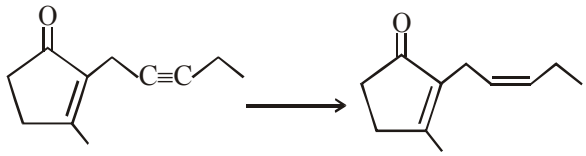
6. 

Product of above reaction will be :

(A) Racemic mixture (B) Diastereomers (C) Meso (D) Constitutional isomers

RD0006

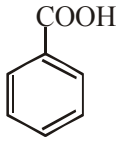
node06\B0BC-BD\Kota\JEE(Advanced)\Enfinite\Chem\Module\Mod.#Halo-Alc.&Eth.-Hydr.-Carb.-Oxi.-Red.-Akl.&Aren.(OC)\Eng\06_Reduction\V0 1_Theory+Ex.p65"

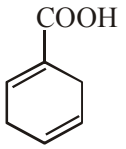
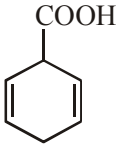
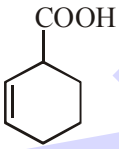
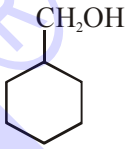
7. 

Which reagent will be used for the above conversion ?

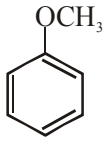
(A) Na/Liq. NH_3 (B) H_2, Pd–$CaCO_3$ (C) Li, Ph–NH_2 (D) H_2, Pt

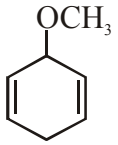
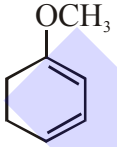
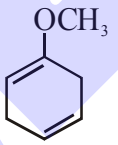
RD0007

8.  $\xrightarrow[\text{Liq.}NH_3]{\text{Li}}$ Major product :

(A)  (B)  (C)  (D) 

RD0008

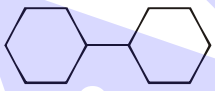
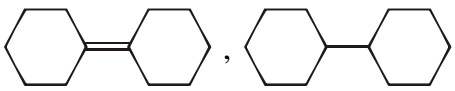
9.  $\xrightarrow{Na,Liq.NH_3}$ Major product :

(A)  (B)  (C)  (D) None of these

RD0009

10.  $\xrightarrow[\Delta]{\text{Conc.}H_2SO_4}$ A $\xrightarrow{Pt/H_2}$ B

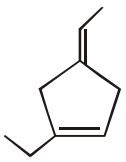
A and B respectively are :

(A) Both  (B) 

(C)  (D) 

RD0010

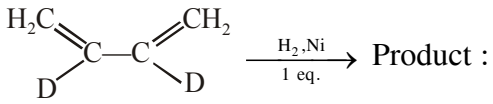
11. If the following compound is treated with Pd in excess of H_2 gas, how many stereoisomers of the product will be obtained ?

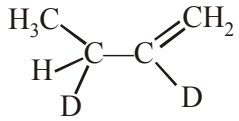


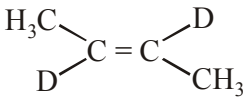
(A) 1 (B) 2 (C) 3 (D) 4

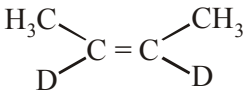
RD0011

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word#\Hdlo-Alc-&-Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&-Ket-(OC)\Eng\06_Reduction\01_Theory+Ex.p65"

12.  Product :

(A) 

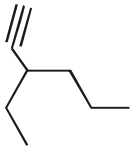
(B) 

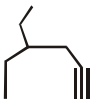
(C) 

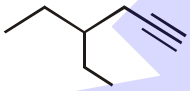
(D) All of these

RD0012

13. Which alkyne gives 3-ethylhexane on catalytic hydrogenation ?

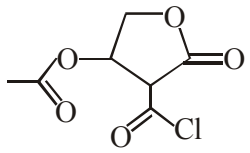
(A) 

(B) 

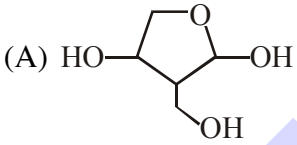
(C) 

(D) All of these

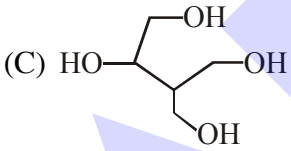
RD0013

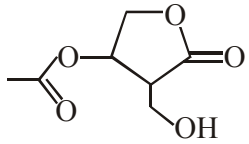
14.  Product

The product is :

(A) 

(B) $CH_3–CH_2–OH$

(C) 

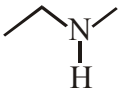
(D) 

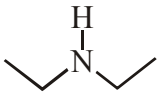
RD0014

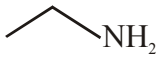
15. 

The final product (B) is :

(A) 

(B) 

(C) 

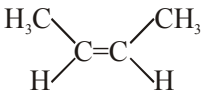
(D) 

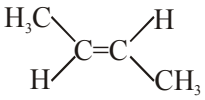
RD0015

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\06_Reduction\V0 1_Theory+Ex.p65"

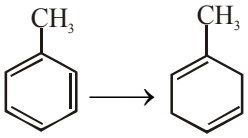
EXERCISE # O-2

1. **Column-I (Conversion)** | **Column-II (Required Reagent)**

(A) $H_3C–C\equiv C–CH_3 \longrightarrow$  (P) H_2 , Pd-$BaSO_4$

(B) $H_3C–C\equiv C–CH_3 \longrightarrow$  (Q) Li , Liq.NH_3

(C) $H_3C–CH=CH–CH_3 \longrightarrow H_3C–CH_2–CH_2–CH_3$ (R) N_2H_4 / H_2O_2 (1 eq.)

(D)  (S) B_2H_6 , CH_3COOH

RD0016

2. $R–\overset{\overset{O}{\|}}{C}–R \xrightarrow{?} R–CH_2–R$

Identify numbers of reagent that can be used for above conversion.

(a) Zn–Hg / HCl (b) $LiAlH_4$

(c) N_2H_4 / $\overset{\ominus}{O}H$ (d) $\begin{matrix}CH_2–SH\\|\\CH_2–SH\end{matrix}$, H_2 / Ni

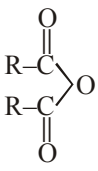
RD0017

3. **Column-I (Conversion)** | **Column-II (Required Reagent)**

(A) $–\overset{\overset{O}{\|}}{C}–OH \longrightarrow –CH_2OH$ (P) $LiAlH_4$

(B) $–\overset{\overset{O}{\|}}{C}–Cl \longrightarrow –CH_2OH$ (Q) $NaBH_4$

(C) $–\overset{\overset{O}{\|}}{C}–OR' \longrightarrow –CH_2OH$ (R) B_2H_6 / THF

(D)  $\longrightarrow R–CH_2OH$ (S) H_2 / Catalyst, Δ

RD0018

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hlde-Alc & Et - Hydr.-Carb.-Oxi.-Red.-Alld & Ket-(OC)\Eng\06_Reduction\V0 1_Theory+Ex.p65"

4. How many alkene on catalytic reduction gives following product.

(i) (A) $\xrightarrow{H_2/Pt}$ n-butane

(ii) (B) $\xrightarrow{H_2/Pt}$ Iso-pentane

(iii) (C) $\xrightarrow{H_2/Pt}$ Neo-pentane

(iv) (D) $\xrightarrow{H_2/Pt}$ Cyclopentane

(v) (E) $\xrightarrow[Pt]{H_2}$ 

RD0019

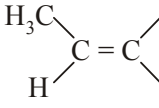
5. Give the expected major product for each reaction, including stereochemistry where applicable.

(a) $CH_3–CH_2–CH=CH_2 \xrightarrow[Pt]{H_2} P_1$

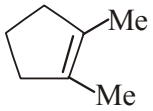
RD0020

(b)  $\xrightarrow[Pt]{D_2} P_2$

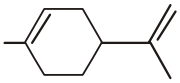
RD0021

(c)  $\xrightarrow[Ni]{D_2} P_3$

RD0022

(d)  $\xrightarrow{Ni/H_2} P_4$

RD0023

(e)  $\xrightarrow[excess]{H_2/Pt} P_5$; $\xrightarrow[1\ eq.]{H_2/Pt} P_6$

RD0024

6. (i) $CH_2=CH–CH_2–CH=O \xrightarrow[excess]{H_2/Pt} P_1$

RD0025

(ii) $CH_2=CH–CH_2–C\equiv N \xrightarrow[excess]{H_2/Pt} P_2$

RD0026

7. (a) Identify the product ?

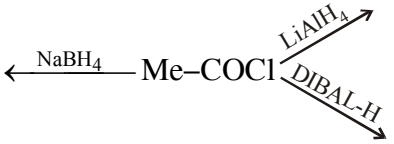
(i) $\xleftarrow{NaBH_4}$ Me–CHO $\xrightarrow[-78°C]{LiAlH_4 \text{ / DIBAL-H}}$

RD0027

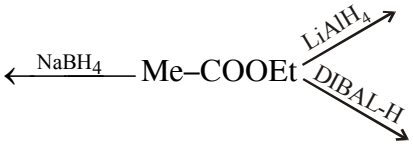
(ii) $\xleftarrow{NaBH_4}$ Me_2CO $\xrightarrow{LiAlH_4 \text{ / DIBAL-H}}$

RD0028

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#\Hdb-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\06_Reduction\V0.1_Theory+Ex.p65"

(iii) 

RD0029

(iv) 

RD0030

(b) Identify the product ?

(v) $\xleftarrow{NaBH_4}$ Me–COOH $\xrightarrow{LiAlH_4}$

RD0031

(vi) $\xleftarrow{NaBH_4}$ Me–COOMe $\xrightarrow{LiAlH_4}$

RD0031

(vii) $\xleftarrow{NaBH_4}$ $Me–CONH_2$ $\xrightarrow{LiAlH_4}$

RD0031

(viii) $\xleftarrow{NaBH_4}$ Me–CONH–Me $\xrightarrow{LiAlH_4}$

RD0032

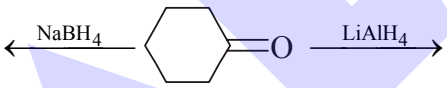
(ix) $\xleftarrow{NaBH_4}$ $Me–CONMe_2$ $\xrightarrow{LiAlH_4}$

RD0032

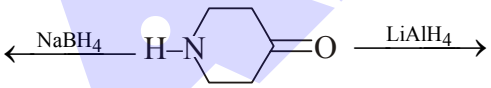
(x) $\xleftarrow{NaBH_4}$ $CH_3–CH = CH_2$ $\xrightarrow{LiAlH_4}$

RD0032

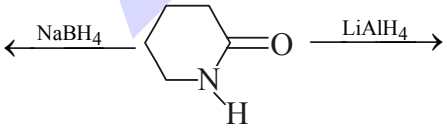
8. Give product in following reactions.

(i) 

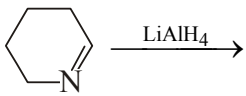
RD0033

(ii) 

RD0034

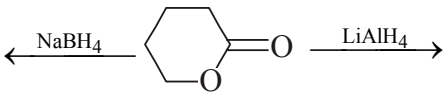
(iii) 

RD0033

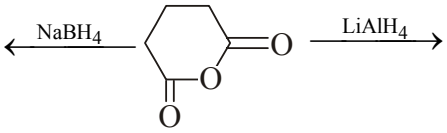
(iv) 

RD0034

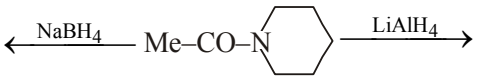
node06\B0BC-BD\Kota VEE\Advanced\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et - Hydr.-Carb.-Oxi.- Red.-Ald & Ket-(OC)\Eng\06_Reduction \01_Theory+Ex.p65"

(v) 

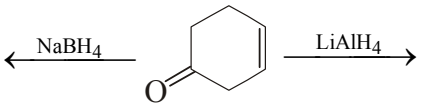
RD0033

(vi) 

RD0034

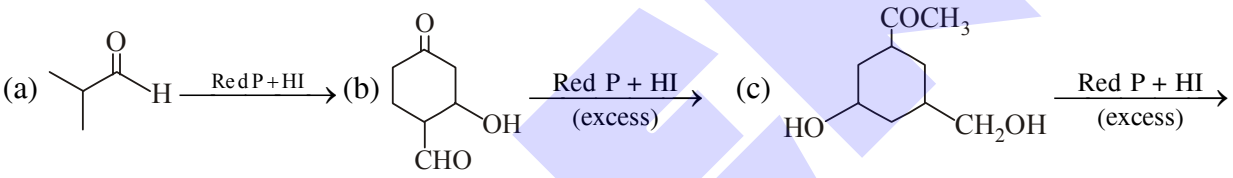
(vii) 

RD0033

(viii) 

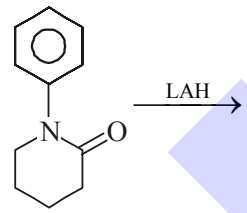
RD0034

9. Give product in following reactions.

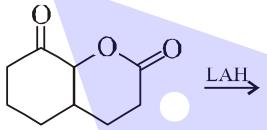


RD0035

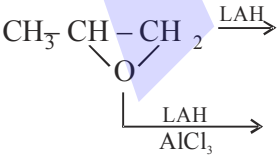
10. Give product in following reactions.

(i) 

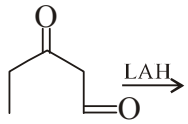
RD0036

(ii) 

RD0037

(iii) 

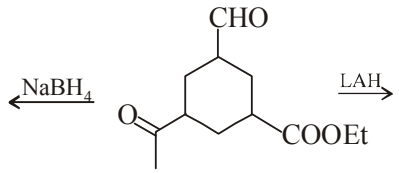
RD0038

(iv) 

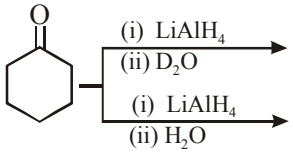
RD0039

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#1\Halo-Alc & Eth -Hydr.-Carb.-Oxi.- Red.- Ald & Ket-(OC)\Eng\06_Reduction\V0 1_Theory+Ex.p65"

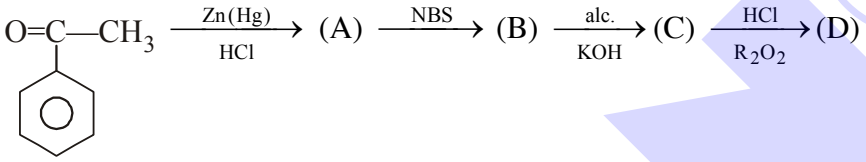
11. Give product in following reactions.

(i) 

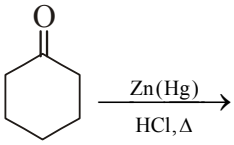
RD0040

(ii) 

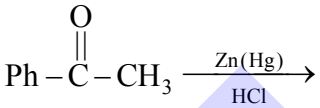
RD0041

12. (a) 

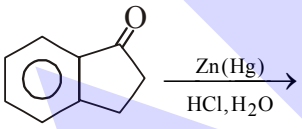
RD0042

13. (a) 

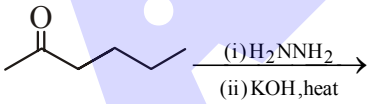
RD0043

(b) 

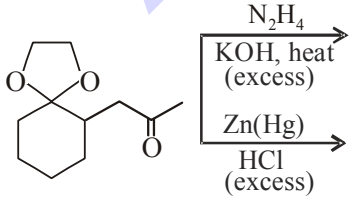
RD0043

(c) 

RD0043

(d) 

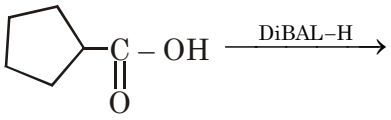
RD0044

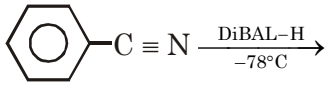
(e) 

RD0045

ALLEN®

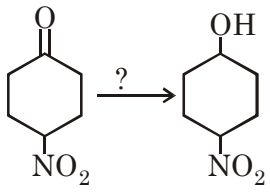
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et - Hydr.-Carb.-Oxi.- Red.-Ald & Ket-(OC)\Eng\06_Reduction\01_Theory+Ex.p65"

14. **(a)** 

(b) 

(c) $CH_3 - \underset{\underset{O}{\|}}{C} - O - CH_2 - CH_3 \xrightarrow[-78^\circ C]{DiBAL-H}$

RD0046

15. 

above conversion can be achieved by

(A) $LiAlH_4$ (B) H_2 / Ni, Δ (C) $NaBH_4$ (D) All

RD0047

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod#\Halo-Alc & Eth -Hydr -Carb -Oxi - Red - Ald & Ket-(OC)\Eng\06_Reduction\V01_Theory+Ex.p65"

EXERCISE # MAIN & ADVANCE

1. Hydrogenation of benzoyl chloride in the presence of Pd and $BaSO_4$ gives : **[IIT 1992]**

(A) Benzyl alcohol (B) Benzaldehyde (C) Benzoic acid (D) Phenol

RD0048

2. Draw the stereochemical structure of the product in the following reactions. **[IIT 1994]**

$$R–C\equiv C–R \xrightarrow[\text{Lindlar catalyst}]{H_2}$$

RD0049

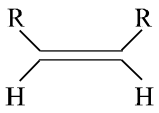
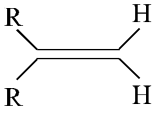
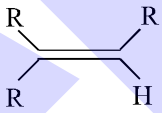
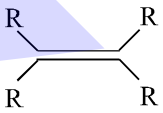
3. Under Wolff Kishner reduction conditions, the conversions which may be brought about is?

(A) Benzaldehyde into Benzyl alcohol (B) Cyclohexanol into Cyclohexane

(C) Cyclohexanone into Cyclohexanol (D) Benzophenone into Diphenylmethane **[IIT 1995]**

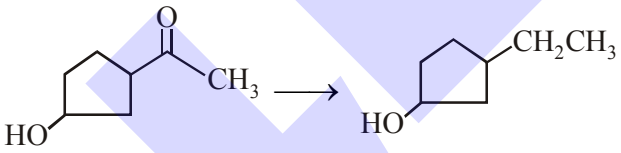
RD0050

4. Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition **[IIT 2000]**

(A)  (B)  (C)  (D) 

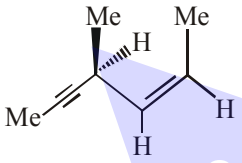
RD0051

5. The appropriate reagent for the following transformation: **[IIT-2000]**



(A) Zn(Hg), HCl (B) NH_2NH_2, OH^- (C) H_2/Ni (D) $NaBH_4$

RD0052

6. 

Hydrogenation of the above compound in the presence of poisoned paladium catalyst gives-

(A) An optically active compound (B) An optically inactive compound **[IIT-2001]**

(C) A racemic mixture (D) A diastereomeric mixture

RD0053

7. When $CH_2=CH–COOH$ is reduced with $LiAlH_4$, the compound obtained will be -**[AIEEE-2003]**

(A) $CH_3–CH_2–CH_2OH$ (B) $CH_3–CH_2–CHO$

(C) $CH_3–CH_2–COOH$ (D) $CH_2=CH–CH_2OH$

RD0054

8. 2–hexyne can be converted into trans–2–hexene by the action of : **[IIT-2004]**

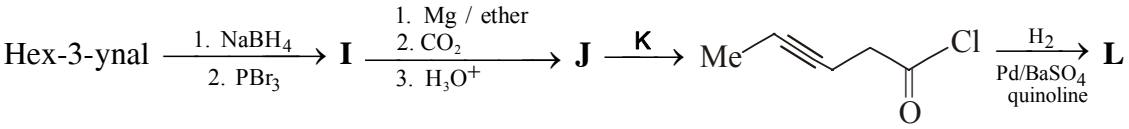
(A) H_2–Pd-$BaSO_4$ (B) Li in liq. NH_3 (C) H_2–PtO_2 (D) $NaBH_4$

RD0055

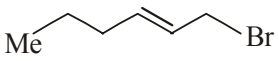
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlc-Alc & Et - Hydr.-Carb.-Oxi.-Red.-Ald & Ket-(OC)\Eng\06_Reduction \V0 1_Theory+Ex.p65"

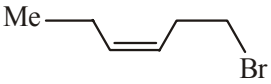
Paragraph for Question No. 09 to 11

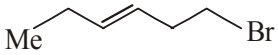
In the following reaction sequence, products **I**, **J** and **L** are formed. **K** represents a reagent.

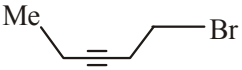


9. The structure of the product **I** is **[JEE-2008]**

(A) 

(B) 

(C) 

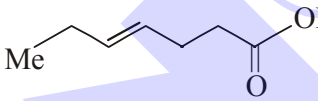
(D) 

RD0056

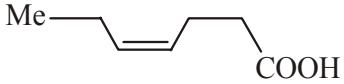
10. The structures of compounds **J** and **K**, respectively, are **[JEE-2008]**

(A) 

and $SOCl_2$

(B) 

and SO_2Cl_2

(C) 

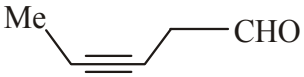
and $SOCl_2$

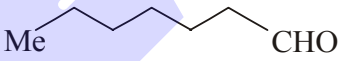
(D) 

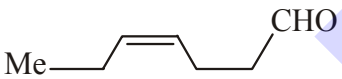
and CH_3SO_2Cl

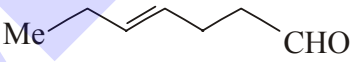
RD0056

11. The structure of product **L** is : **[JEE-2008]**

(A) 

(B) 

(C) 

(D) 

RD0056

12. Consider all possible isomeric ketones including stereoisomers of MW = 100, All these isomers are independently reacted with $NaBH_4$ (**NOTE :** stereoisomers are also reacted separately). The total number of ketones that give a racemic product(s) is/are. **[JEE-2014]**

RD0057

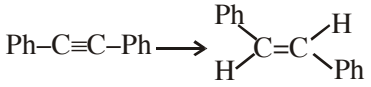
13. In the reaction, **[JEE-2014]**

$CH_3COOH \xrightarrow{LiAlH_4} A \xrightarrow{PCl_5} B \xrightarrow{AlC.\,KOH} C$, Final product C is.

(A) Ethylene (B) Acetyl chloride (C) Acetaldehyde (D) Acetylene

RD0058

14. The reagent needed for converting is : **[JEE-2014]**



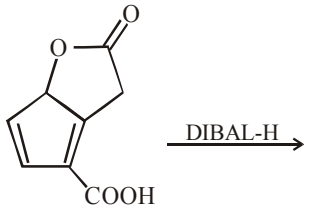
(A) H_2 / Lindlar Cat. (B) Cat. hydrogenation

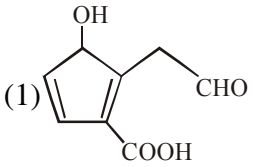
(C) $LiAlH_4$ (D) Li / NH_3

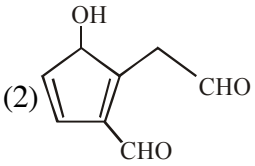
RD0059

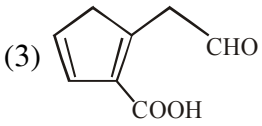
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod#Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Akl.&Ren.(OC)\Eng\06_Reduction\V01_Theory+Ex.p65"

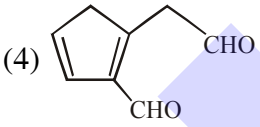
15. The major product obtained in the following reaction is : **[Jee Main-2017]**



(1) 

(2) 

(3) 

(4) 

RD0060

16. Glucose on prolonged heating with HI gives : **[Jee Main - 2018]**

(1) 1–Hexene (2) Hexanoic acid (3) 6-iodohexanal (4) n-Hexane

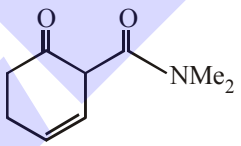
RD0061

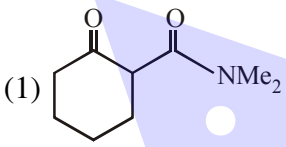
17. The *trans*-alkenes are formed by the reduction of alkynes with : **[Jee Main - 2018]**

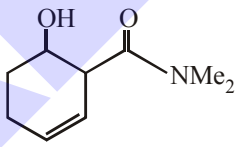
(1) $NaBH_4$ (2) Na/liq.NH_3 (3) Sn–HCl (4) H_2–Pd/C, $BaSO_4$

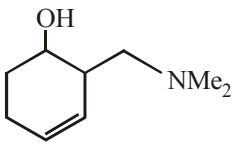
RD0062

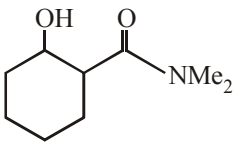
18. The main reduction product of the following compound with $NaBH_4$ in methanol is :- **[Jee Main-Online Morning 2018]**



(1) 

(2) 

(3) 

(4) 

RD0063

19. The major product of following reaction is : **[Jee-Main Online-(Jan)-2019]**

$R-C\equiv N \xrightarrow[(2)H_2O]{(1)AlH(i-Bu_2)} ?$

(1) RCHO (2) RCOOH (3) RCH_2NH_2 (4) $RCONH_2$

RD0064

20. Wilkinson catalyst is : **[Jee-Main Online-(Jan)-2019]**

(1) $[(Ph_3P)_3RhCl]$ $(Et = C_2H_5)$ (2) $[Et_3P)_3IrCl]$

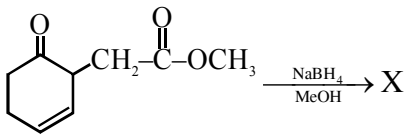
(3) $[Et_3P)_3RhCl]$ (4) $[Ph_3P)_3IrCl]$

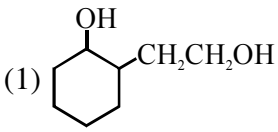
RD0065

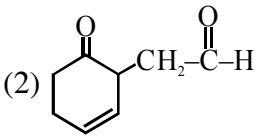
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hlde-Alc & Et - Hydr.-Carb.-Oxi.-Red.-Ald & Ket-(OC)\Eng\06_Reduction \V0 1_Theory+Ex.p65"

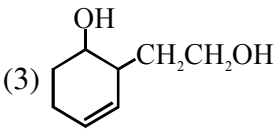
21. The major product 'X' formed in the following reaction is :

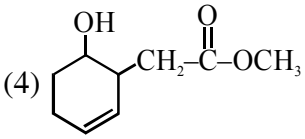
[JEE-MAIN-On-line-(Jan)-2019]



(1) 

(2) 

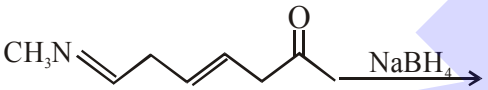
(3) 

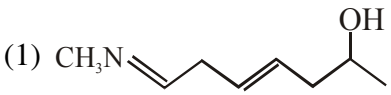
(4) 

RD0066

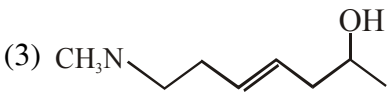
22. The major product of the following recation is:

[JEE-MAIN-On-line-(Jan)-2019]



(1) 

(2) 

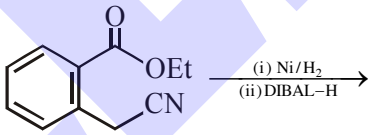
(3) 

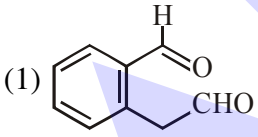
(4) 

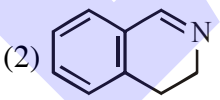
RD0067

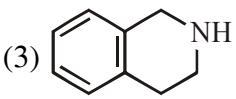
23. The major product of the following reaction is:

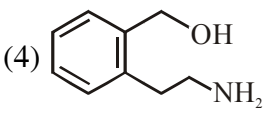
[JEE-MAIN-On-line-(Jan)-2019]



(1) 

(2) 

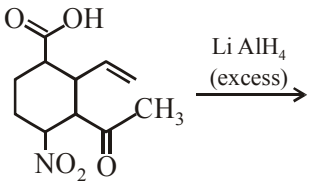
(3) 

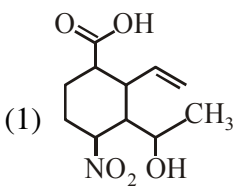
(4) 

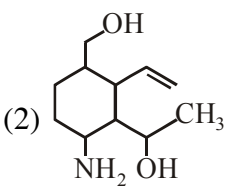
RD0068

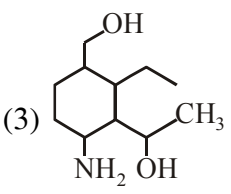
24. The major product obtained in the following reaction is :-

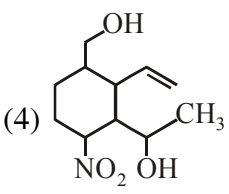
[JEE-MAIN-On-line-(Jan)-2019]



(1) 

(2) 

(3) 

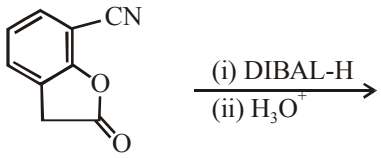
(4) 

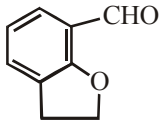
RD0069

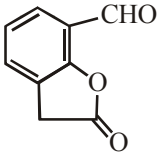
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\06_Reduction\01_Theory+Ex.p65"

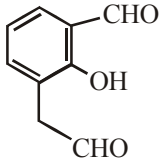
25. The major product of the following reaction is:

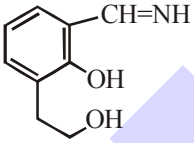
[JEE-MAIN-On-line-(Jan)-2019]



(1) 

(2) 

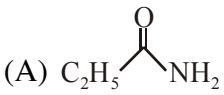
(3) 

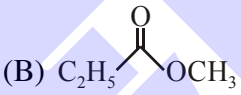
(4) 

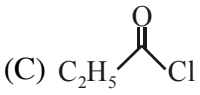
RD0070

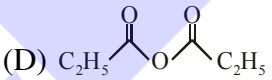
26. The increasing order of the reactivity of the following with $LiAlH_4$ is :

[JEE-MAIN-On-line-(Jan)-2019]

(A) 

(B) 

(C) 

(D) 

(1) (A) < (B) < (D) < (C)

(2) (A) < (B) < (C) < (D)

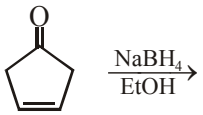
(3) (B) < (A) < (D) < (C)

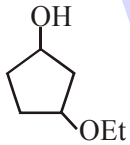
(4) (B) < (A) < (C) < (D)

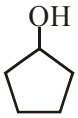
RD0071

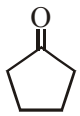
27. The major product of the following reaction is:

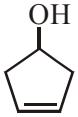
[JEE-MAIN-On-line-(Jan)-2019]



(1) 

(2) 

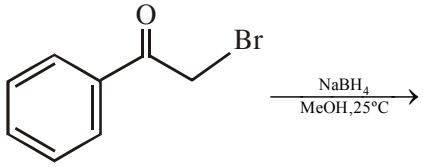
(3) 

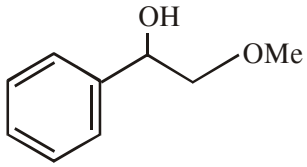
(4) 

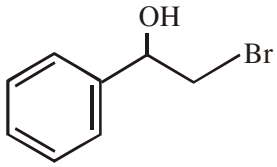
RD0072

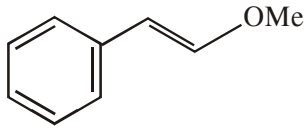
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word#\Hlkb-Alc-&-Et.-Hydr.-Carb.-Oxi.-Red.-Ald.-&-Ket.-(OC)\Eng\06_Reduction\01_Theory+Ex.p65"

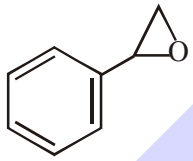
28. The major product of the following reaction is: **[JEE-MAIN-On-line-(April)-2019]**



(1) 

(2) 

(3) 

(4) 

RD0073

29. The major product of the following reaction is : **[JEE-MAIN-On-line-(April)-2019]**

$CH_3CH=CHCO_2CH_3 \xrightarrow{LiAlH_4}$

(1) $CH_3CH_2CH_2CHO$ (2) $CH_3CH=CHCH_2OH$

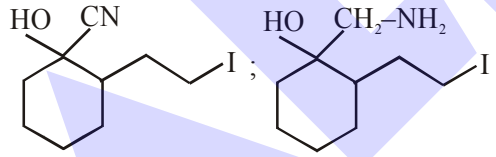
(3) $CH_3CH_2CH_2CO_2CH_3$ (4) $CH_3CH_2CH_2CH_2OH$

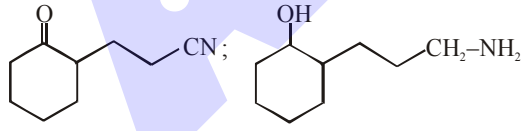
RD0074

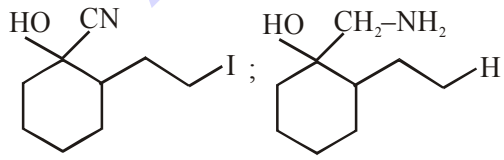
30. The major products A and B for the following reactions are, respectively:

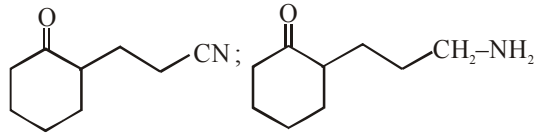


[JEE-MAIN-On-line-(April)-2019]

(1) 

(2) 

(3) 

(4) 

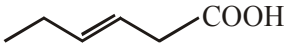
RD0075

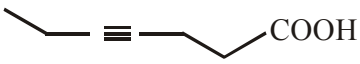
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\06_Reduction\V01_Theory+Ex.p65"

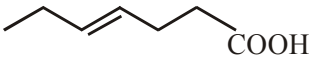
31. What is the product of following reaction ? **[JEE-MAIN-On-line-(Jan)-2020]**

Hex-3-ynal $\xrightarrow[\text{(ii) } PBr_3 \\ \text{(iii) Mg/ether} \\ \text{(iv) } CO_2/H_3O^+]{\text{(i) } NaBH_4}$?

(1) 

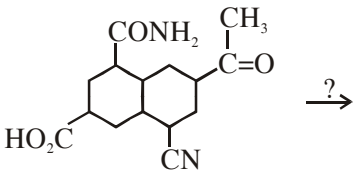
(2) 

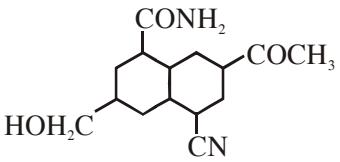
(3) 

(4) 

RD0076

32. The most suitable reagent for the given conversion is : **[JEE-MAIN-On-line-(Jan)-2020]**





(1) $LiAlH_4$ (2) $NaBH_4$ (3) H_2/Pd (4) B_2H_6

RD0077

33. **Statement I :** Sodium hydride can be used as an oxidising agent. **[JEE-MAIN-2021]**

Statement II : The lone pair of electrons on nitrogen in pyridine makes it basic.

Choose the CORRECT answer from the options given below :

(1) Both statement I and statement II are false

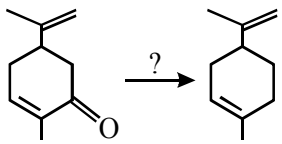
(2) Statement I is true but statement II is false

(3) Statement I is false but statement II is true

(4) Both statement I and statement II are true

RD0078

34. **[JEE-MAIN-2021]**

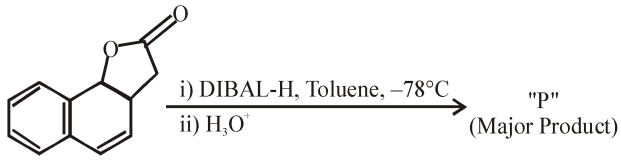


Which of the following reagent is suitable for the preparation of the product in the above reaction?

(1) $NaBH_4$ (2) $NH_2–NH_2/C_2H_5O^{\ominus}Na^{\oplus}$ (3) Ni/H_2 (4) Red P + Cl_2

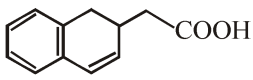
RD0079

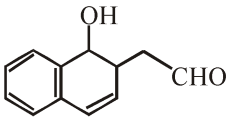
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hlde-Alc & Et - Hydr.-Carb.-Oxi.- Red.-Ald & Ket-(OC)\Eng\06_Reduction\V0.1_Theory+Ex.p65"

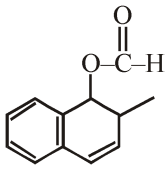
35. 

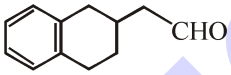
[JEE-MAIN-2021]

The product "P" in the above reaction is :

(1) 

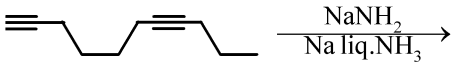
(2) 

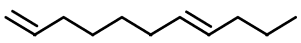
(3) 

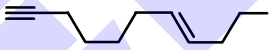
(4) 

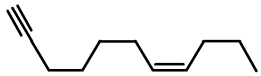
RD0080

36. The major product formed in the following reaction is **[JEE ADVANCE-2021]**



(A) 

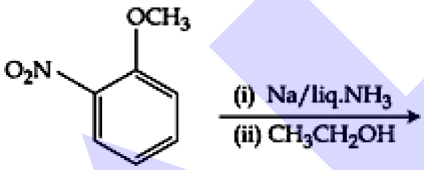
(B) 

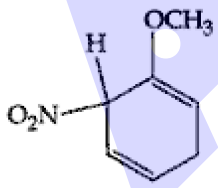
(C) 

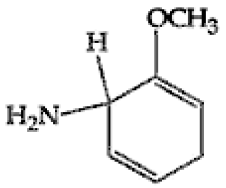
(D) 

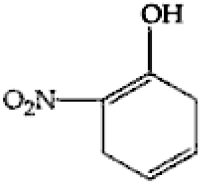
RD0081

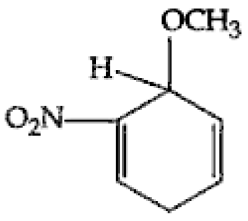
37. The major product of the following reaction is **[JEE-MAIN-2022]**



(1) 

(2) 

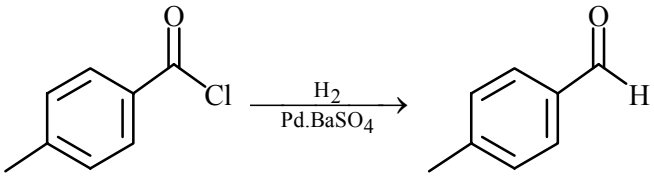
(3) 

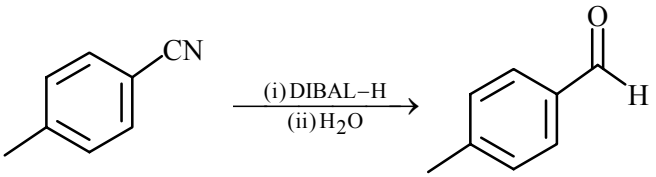
(4) 

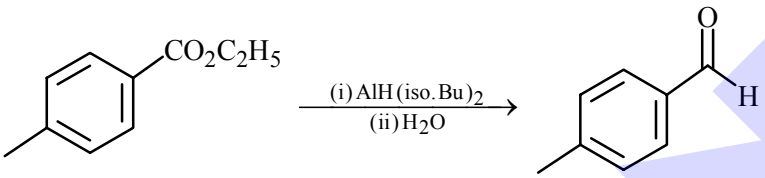
RD0082

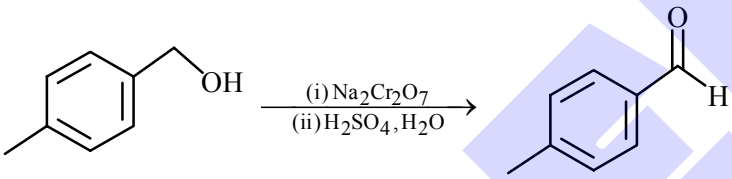
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc & Et- Hydr.-Carb.-Oxi.-Red.-Ald & Ket-(OC)\Eng\06_Reduction\01_Theory+Ex.p65"

38. Which one of the following reactions does not represent correct combination of substrate and product under the given conditions ? **[JEE-MAIN-2022]**

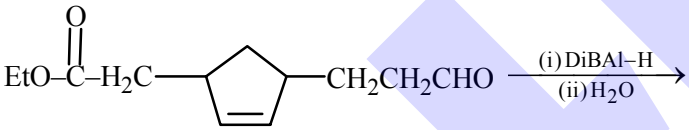
(1) 

(2) 

(3) 

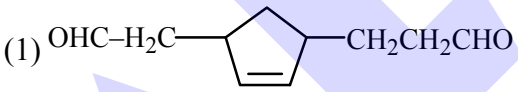
(4) 

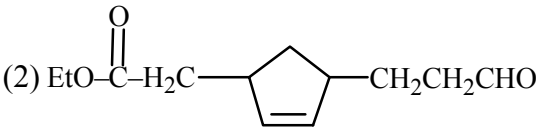
RD0083

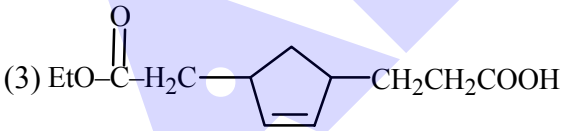
39. 

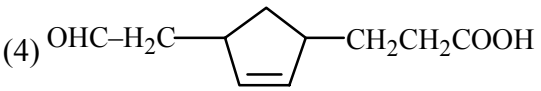
[JEE-MAIN-2022]

Consider the above reaction and predict the major product.

(1) 

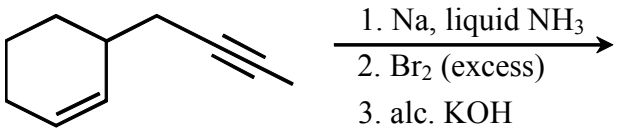
(2) 

(3) 

(4) 

RD0084

40. The number of isomeric tetraenes (NOT containing sp-hybridized carbon atoms) that can be formed from the following reaction sequence is _____. **[JEE ADVANCE-2022]**



RD0085

node06\B09C-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Halo-Alc & Et.-Hydr.-Carb.-Oxi.-Red. Ald & Ket\OC\Eng\06_Reduction \01_Theory+Ex.p65"

ANSWER KEY

EXERCISE # O-1

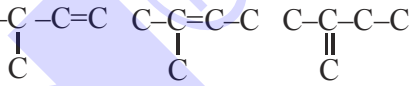
Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	C	C	A	C	B	B	C	B
Que.	**11**	**12**	**13**	**14**	**15**					
Ans.	C	C	D	D	B					

EXERCISE # O-2

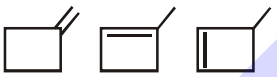
1. **(A)→P, R, S; (B)→Q; (C)→R, S; (D)→Q**

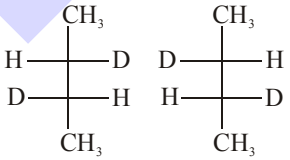
2. **(3)**

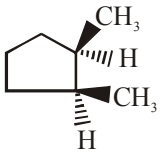
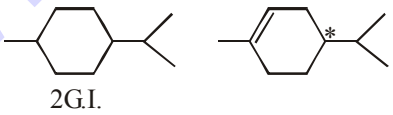
3. **(A)→P, R, S; (B)→P, Q, S; (C)→P, R, S; (D)→P, R, S**

4. (i) Cis & trans 2-butene & 1-butene ; (ii) 

(iii) Zero (Neo-pentane can not be prepared by catalytic hydrogenation of alkene) ; (iv) One

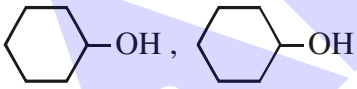
(v) Including optical = 4  , Excluding optical = 3.

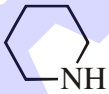
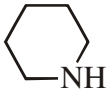
5. (a)  (b)  (c) 

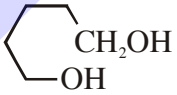
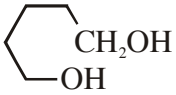
(d)  (e) 

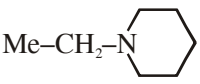
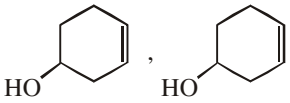
6. (i) $P_1 = CH_3–CH_2–CH_2–CH_2–OH$ (ii) $P_2 = CH_3–CH_2–CH_2–CH_2–NH_2$

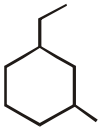
8.

(i)  (ii) 

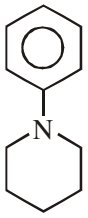
(iii) No reaction,  (iv) 

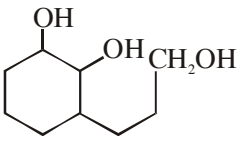
(v) No reaction,  (vi) No reaction, 

(vii) No reaction,  (viii) 

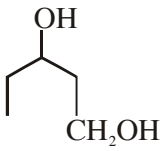
9. (a)  (b)  (c) 

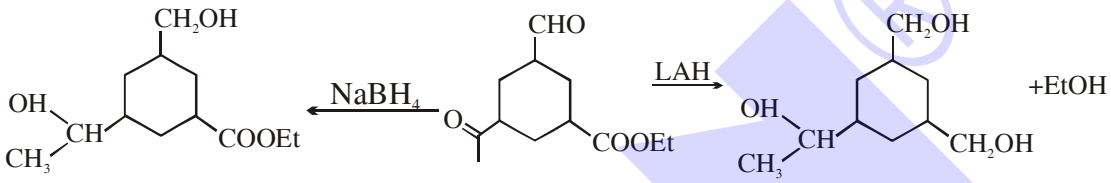
node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod.#Halo-Alc.&Et.-Hydr.-Carb.-Oxi.-Red.-Ald.&Ket-(OC)\Eng\06_Reduction\V0 1_Theory+Ex.p65"

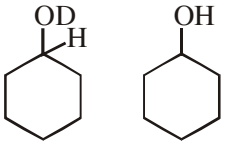
10. (i) 

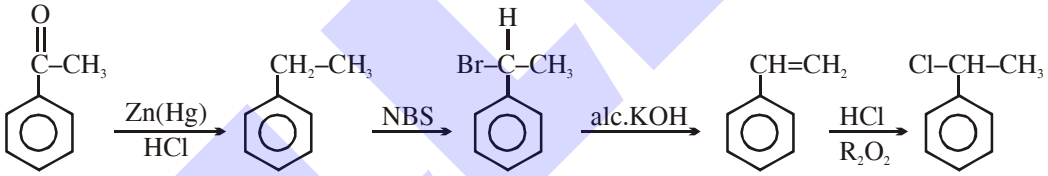
(ii) 

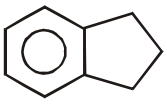
(iii) $CH_3–\underset{\underset{OH}{|}}{CH}–CH_3$, $CH_3–CH_2–CH_2OH$

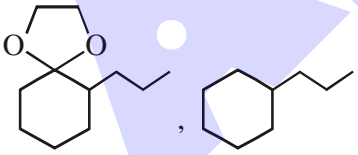
(iv) 

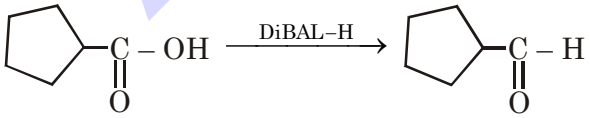
11. (i) 

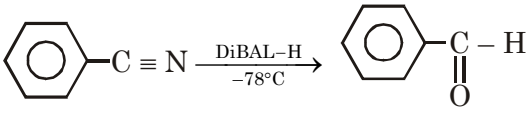
(ii) 

12. (a) 

13. (a)  (b) $Ph–CH_2–CH_3$ (c)  (d) 

(e) 

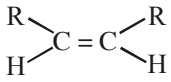
14. **(a)** 

(b)  **(c)** $CH_3 - CHO$

15. **(C)**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word#\Hlde-Alc-&-Et.-Hydr.-Carb.-Oxi.-Red.-Ald.-&-Ket.-(OC)\Eng\06_Reduction\01_Theory+Ex.p65"

EXERCISE # (MAIN & ADVANCE)

1. **(B)** **2.**  **3.** **(D)** **4.** **(A)**

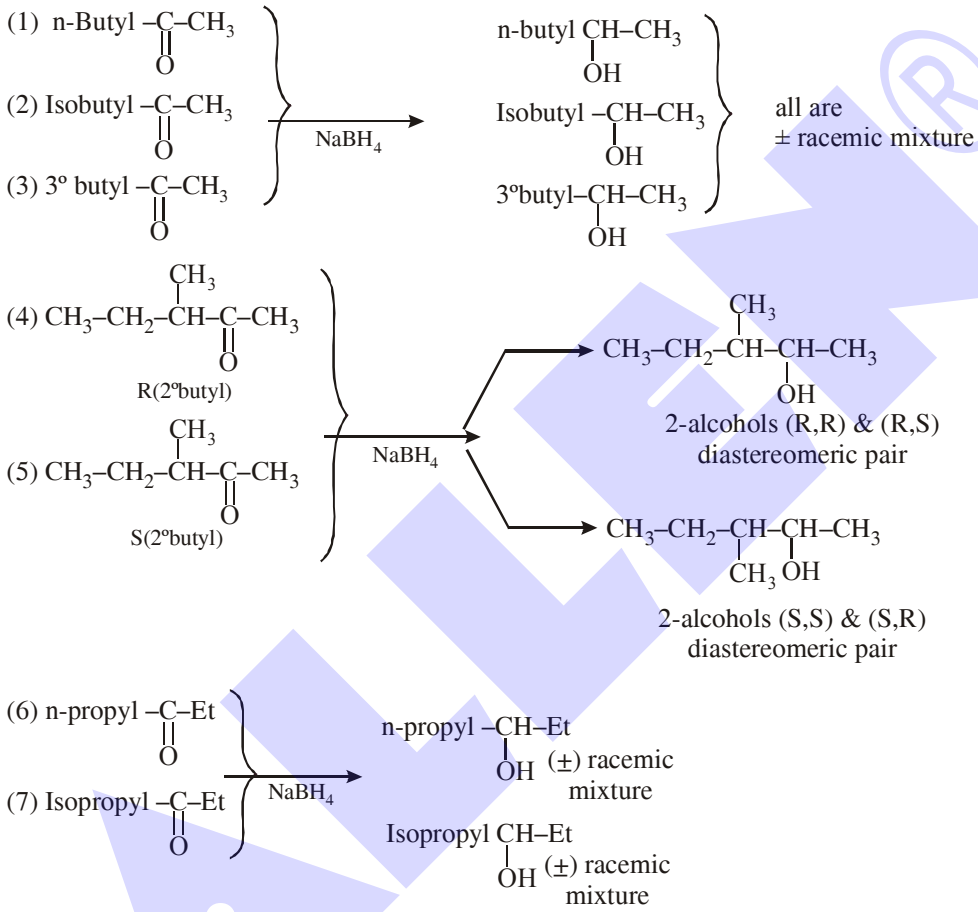
5. **(B)** **6.** **(B)** **7.** **(D)** **8.** **(B)**

9. **(D)** **10.** **(A)** **11.** **(C)**

12. **(5)**

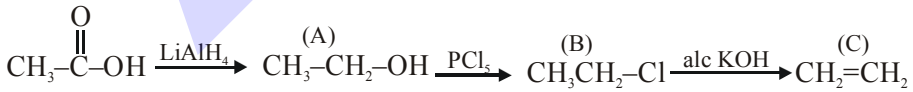
M. wt 100 of ketone

So m. formula = $C_6H_{12}O$



(1 ; 2 ; 3 ; 6 ; 7)

13. (A)



Reduction of CH_3COOH will produce $CH_3–CH_2–OH$(A)

Now reaction of $CH_3–CH_2–OH$(A) with PCl_5 will produce $CH_3–CH_2–Cl$(B).

Now alcoholic KOH with Et–Cl will produce $CH_2=CH_2$(C) Ethylene by E_2 elemination.

Thus is (1) Ethylene.

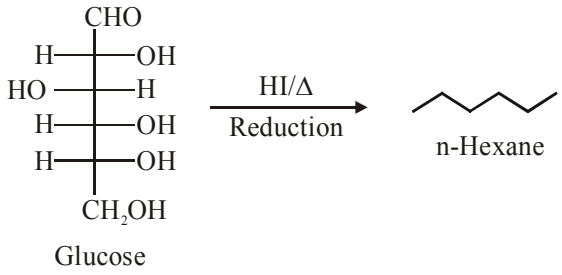
14. **(D)**

15. **(1)**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Mod#Hdb-Alc&Et-Hydr-Carb-Oxi-Red-Ald&Ket-(OC)\Eng\06_Reduction\V0 1_Theory+Ex.p65"

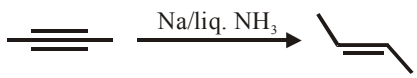
16. (4)

Sol.



17. (2)

Sol.



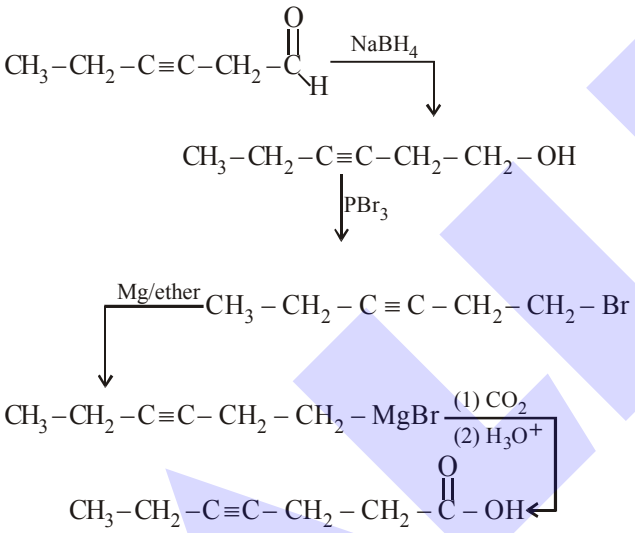
Birch reduction is anti addition. So trans alkene will be produced.

18.	**(2)**	**19.**	**(1)**	**20.**	**(1)**	**21.**	**(4)**
22.	**(3)**	**23.**	**(2)**	**24.**	**(2)**	**25.**	**(3)**
26.	**(1)**	**27.**	**(4)**	**28.**	**(4)**	**29.**	**(2)**

30. (2)

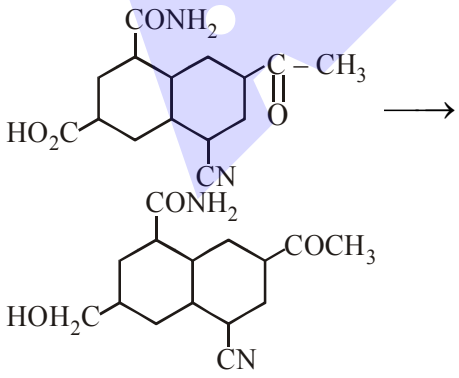
31. (3)

Sol.



32. (4)

Sol.



Most suitable reagent for given conversion is B_2H_6 (electrophilic reducing agent)

33.	**(3)**	**34.**	**(2)**	**35.**	**(2)**	**36.**	**(B)**
37.	**(1)**	**38.**	**(4)**	**39.**	**(1)**	**40.**	**(2)**

node06\B0BC-BD\Kota\JEE(Advanced)\Enthuse\Chem\Module\Word #\Hdlo-Alc & Et - Hydr.-Carb.-Oxi.-Red.-Ald & Ket-(OC)\Eng\06_Reduction\V0 1_Theory+Ex.p65"